EXNO: 5 Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides

Test Case 1:

You are experimenting with a rectangular waveguide to analyze the relationship between its dimensions and wave propagation. The waveguide dimensions are a=4.0 cm, b=2.0 cm and the operating frequency is f=8 GHz.

1. Calculate the cutoff frequencies for TE10modes.

2. Predict the phase and group velocities for the TE10mode at the given frequency.

3. If the width of the waveguide is reduced to a=3.0 cm, how does it affect the cutoff frequency and propagation characteristics?

AIM: To find the group velocity, phase velocity and phase constant of Electromagnetic waves in Rectangular Waveguides whose dimensions are a=4.0 cm, b=2.0 cm and the operating frequency is f=8 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

PROGRAM:

```
clc;

clear;

close;

//TE10 Mode

a=8.636*10^-2;

b=4.318*10^-2;

f=4*10^9;

V=3*10^8;
```

m=input("enter the value of m= "); n=input("enter the value of n= ");

fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);

disp("Cut off frquency=",fc*10^-9);

//let phase velocity = A

A=V/sqrt(1-(fc/f)^2);

disp("Phase velocity in Mm/sec=",A*10^

//let group velocity = B

B=V*V/A;

disp("Group velocity in Mm/sec =", B*10^

m= "); n=input("enter the value of n= ");

fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2); 9);

disp("Phase velocity in Mm/sec=",A*10^-6);

sp("Group velocity in Mm/sec =", B*10^-6);

OUTPUT:

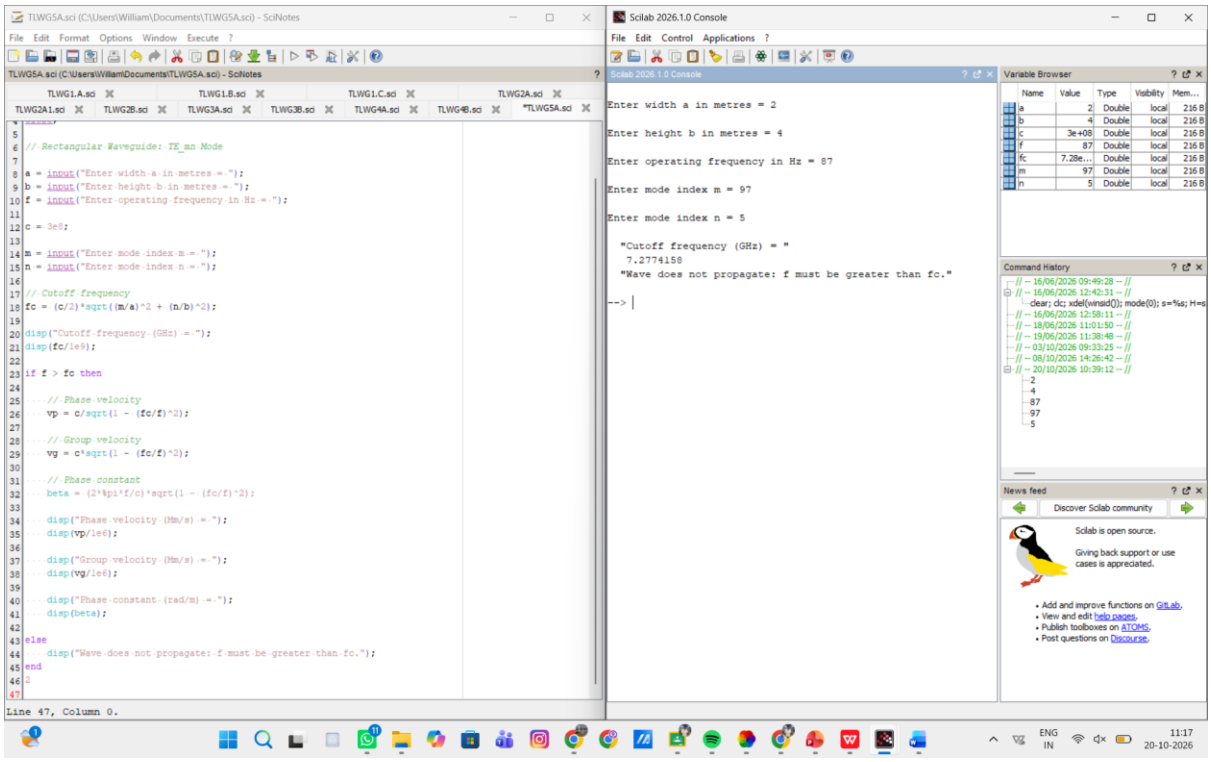
Case 1:

Original Dimensions (a=4.0 cm,b=2.0 cm)

Inputs:

- m=1, n=0.
- f=8 GHz.

Width \(a\)	**\(f_c\) (GHz)**	**\(v_p\) (Mm/s)**	**\(v_g\) (Mm/s)**
4.0 cm	3.750	339.624	264.999
3.5 cm	4.286	355.282	253.320
3.0 cm	5.000	384.308	234.187
2.5 cm	6.000	453.557	198.431
2.0 cm	7.500	862.105	104.396



Test Case 2 :

In a radar system operating at =15 GHz the waveguide dimensions are = 2.5 and □ = 1.25 cm

1. Compute the cutoff frequencies for TE10 modes.

2. Determine the phase and group velocities for the TE10 mode.

3. If dispersion is critical to radar signal accuracy, what strategies would you recommend

maintaining signal fidelity?

AIM: To find the group velocity, phase velocity and phase constant of Electromagnetic waves in

Rectangular Waveguides whose dimensions are a= 2.5 cm, b=1.25 cmand the operating frequency is

f=15 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

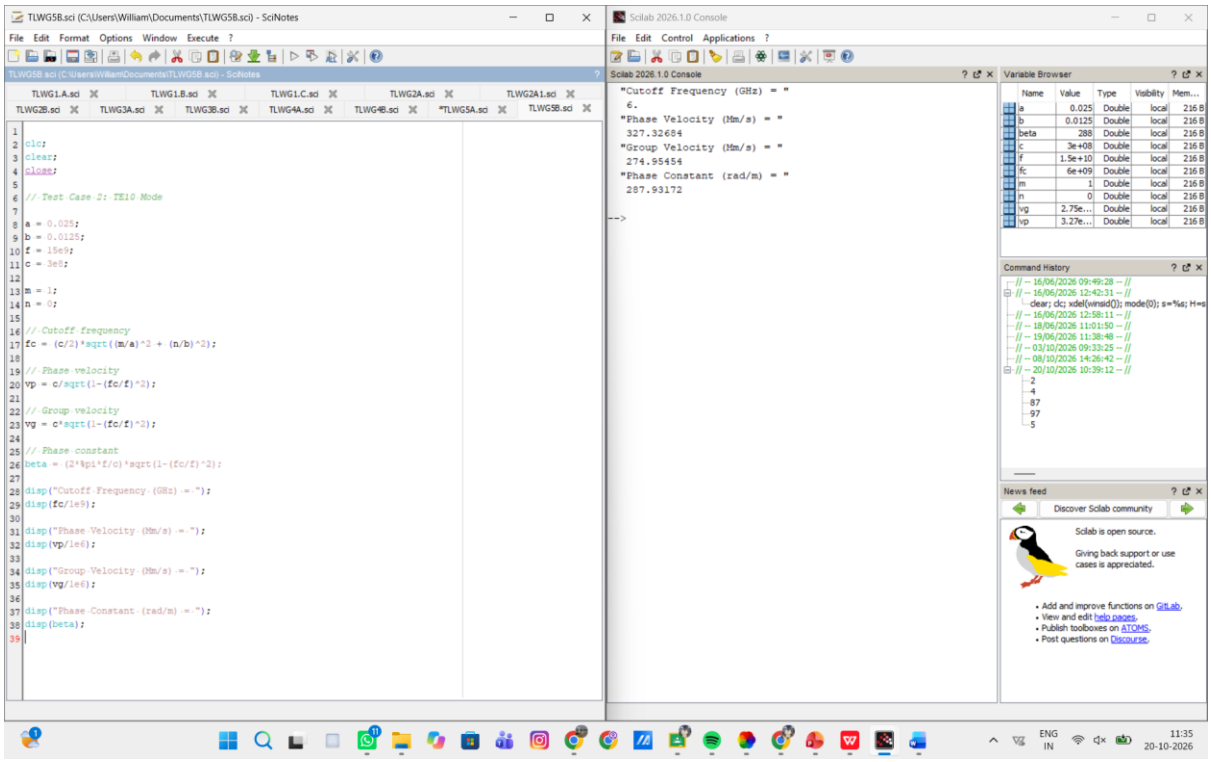
PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= "); n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

Table 2: TE\(_{10}\) mode results

Parameter	**Simulation Results**	**Theoretical Results**
Width \(a\)	0.025 m	0.025 m
Height \(b\)	0.0125 m	0.0125 m
Operating frequency	15 GHz	15 GHz

Cutoff frequency \(f_c\)	6 GHz	6 GHz
Phase velocity \(v_p\)	327.327 Mm/s	327.327 Mm/s
Group velocity \(v_g\)	274.955 Mm/s	274.955 Mm/s
Phase constant \(\beta\)	287.932 rad/m	287.932 rad/m
Propagation	Yes	Yes



RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

Test Case 3 :

A waveguide with dimensions a=5.0 cm and b=2.5 cmis used to carry a signal at f=10 GHz

1. Determine the number of TE modes that can propagate at this frequency.

2. For the TE10mode, calculate the phase velocity and group velocity.

3. Discuss how the presence of multiple propagating modes affects the transmission quality and suggest practical solutions to limit multimode effects.

AIM:

To find the group velocity, phase velocity and phase constant of Electromagnetic waves in

Rectangular Waveguides whose dimensions are a= 5.0 cm, b=2.5 cmand the operating frequency

is f=10 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
```

disp("Phase velocity in Mm/sec=",A*10^-6);

//let group velocity = B

B=V*V/A;

disp("Group velocity in Mm/sec =", B*10^-6);

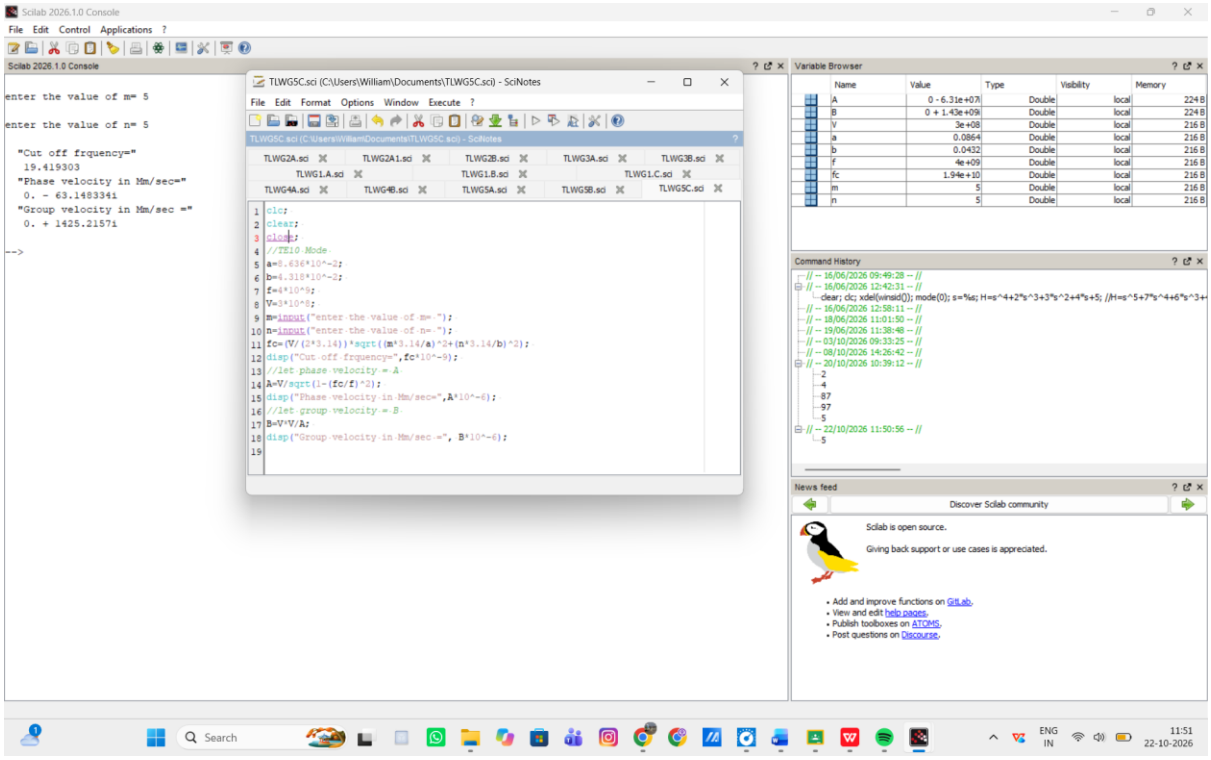
Table 3: TE\(_{10}\) mode results

Parameter	Simulation Results	Theoretical Results
Width \(a\)	0.05 m	0.05 m
Height \(b\)	0.025 m	0.025 m
Operating frequency	10 GHz	10 GHz
Cutoff frequency \(f_c\)	3 GHz	3 GHz
Phase velocity \(v_p\)	314.485 Mm/s	314.485 Mm/s
Group velocity \(v_g\)	286.182 Mm/s	286.182 Mm/s
Phase constant \(\beta\)	199.793 rad/m	199.793 rad/m
Propagation	Yes	Yes

Table 2: Cutoff frequency and propagation of TE modes

Mode	Cutoff frequency (GHz)	Propagation
TE\(_{10}\)	3.000	Yes
TE\(_{01}\)	6.000	Yes
TE\(_{11}\)	6.708	Yes
TE\(_{20}\)	6.000	Yes
TE\(_{21}\)	8.485	Yes
TE\(_{02}\)	12.000	No
TE\(_{30}\)	9.000	Yes

Result: Six TE modes can propagate at 10 GHz: TE\(_{10}\), TE\(_{01}\), TE\(_{11}\), TE\(_{20}\), TE\(_{21}\) and TE\(_{30}\).



RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.

Test Case 4 :

A research lab is experimenting with high-order TE modes (m,n>1) in a waveguide with dimensions a=3.0 cm b=1.5 cm, and an operating frequency of f=12 GHz.

1. Calculate the cutoff frequencies for TE10 mode.

2. Compute the phase and group velocities of this mode.

3. Discuss how high-order modes can be utilized for advanced applications such as mode multiplexing in communications.

AIM:

To find the group velocity, phase velocity and phase constant of Electromagnetic waves in

Rectangular Waveguides whose dimensions are a=3.0 cm, b=1.5 cmand the operating frequency

is f=12 GHz.

SOFTWARE REQUIRED:

SCILAB Version 6.1.1

PROCEDURE:

● Open SCILAB software

● Open file-new

● Put the input values

● Type the program

● Execute the program

● End the program

PROGRAM:

```
clc;
clear;
close;
//TE10 Mode
a=8.636*10^-2;
b=4.318*10^-2;
f=4*10^9;
V=3*10^8;
m=input("enter the value of m= ");
n=input("enter the value of n= ");
fc=(V/(2*3.14))*sqrt((m*3.14/a)^2+(n*3.14/b)^2);
disp("Cut off frquency=",fc*10^-9);
//let phase velocity = A
A=V/sqrt(1-(fc/f)^2);
disp("Phase velocity in Mm/sec=",A*10^-6);
//let group velocity = B
B=V*V/A;
disp("Group velocity in Mm/sec =", B*10^-6);
```

OUTPUT:

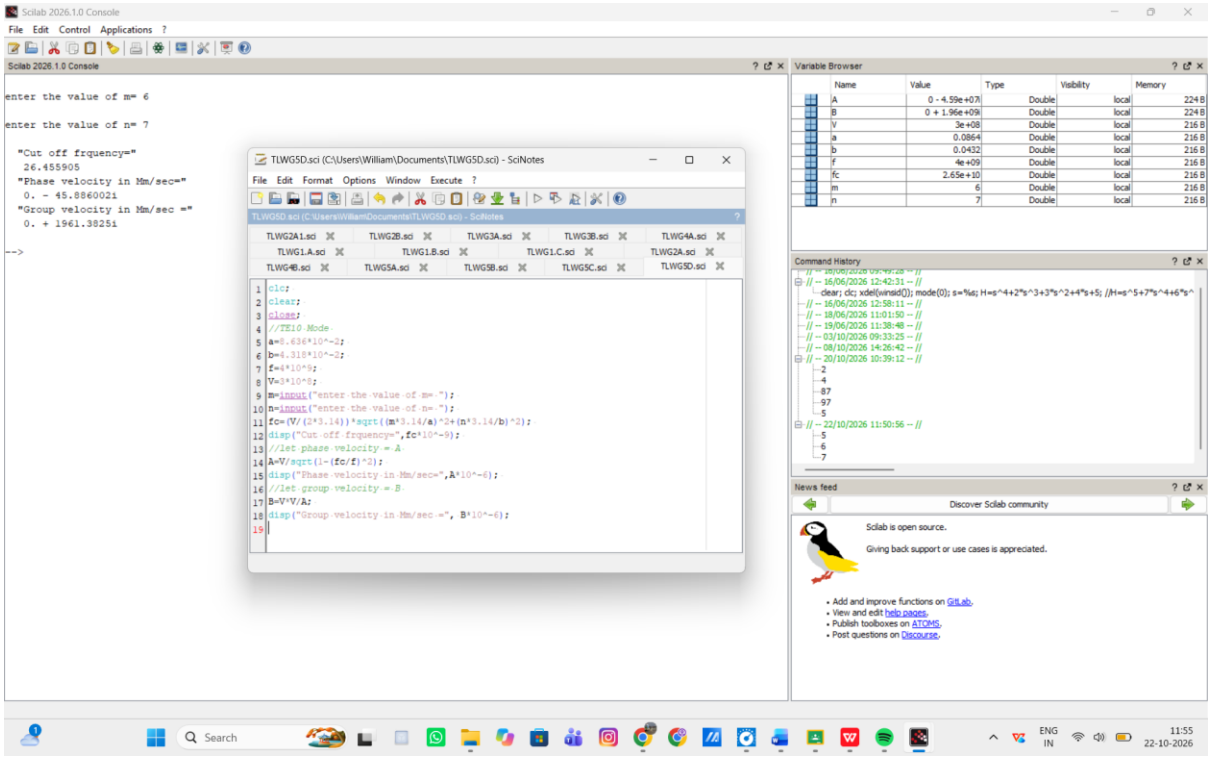
Waveguide Parameters:

 Width (a) = 3.0 cm=0.03 m

 Height (b) = 1.5 cm=0.015 m

 Operating Frequency (f) = 12 GHz=12×109 Hz

 Speed of Light (V) = 3×108m/s



TEST CASE 4

Given: \(a=3.0\text{ cm}\), \(b=1.5\text{ cm}\), \(f=12\text{ GHz}\).

Table 4: Cutoff frequency and propagation of TE modes

Mode	Cutoff frequency (GHz)	Propagation
TE\(_{10}\)	5.000	Yes
TE\(_{20}\)	10.000	Yes
TE\(_{01}\)	10.000	Yes
TE\(_{11}\)	11.180	Yes
TE\(_{21}\)	14.142	No
TE\(_{30}\)	15.000	No

Table 5: TE\(_{10}\) mode results

Parameter	Simulation Results	Theoretical Results
Width \(a\)	0.03 m	0.03 m
Height \(b\)	0.015 m	0.015 m
Operating frequency	12 GHz	12 GHz
Cutoff frequency \(f_c\)	5 GHz	5 GHz
Phase velocity \(v_p\)	330.011 Mm/s	330.011 Mm/s
Group velocity \(v_g\)	272.718 Mm/s	272.718 Mm/s
Phase constant \(\beta\)	228.472 rad/m	228.472 rad/m
Propagation	Yes	Yes

Summary of all test cases

Test case	Operating frequency	Cutoff frequency (TE\(_{10}\))	Phase velocity (Mm/s)	Group velocity (Mm/s)
2	15 GHz	6 GHz	327.327	274.955
3	10 GHz	3 GHz	314.485	286.182
4	12 GHz	5 GHz	330.011	272.718

RESULT:

The group velocity, phase velocity and phase constant of waves in a rectangular waveguide are estimated using SCILAB.